

1. In logistic regression given $\mathbf{x}$ and parameters $w \in \mathbb{R}^{n_x}$, $b \in \mathbb{R}$. Which of the following best expresses what we want $\hat{y}$ to tell us? 1 point

- ☒ $P(y = 1|\mathbf{x})$
☐ $P(y = \hat{y}|\mathbf{x})$
☐ $\sigma(W \mathbf{x})$
☐ $\sigma(W \mathbf{x} + b)$

2. Suppose that $\hat{y} = 0.9$ and $y = 1$. What is the value of the "Logistic Loss"? Choose the best option. 1 point

- ☐ 0.005
☐ 0.105
☐ $+\infty$
☒ $\mathcal{L}(\hat{y}, y) = -(\hat{y} \log y + (1 - \hat{y}) \log(1 - y))$

3. Suppose `img` is a (32,32,3) array, representing a 32x32 image with 3 color channels red, green and blue. How do you reshape this into a column vector `x`? 1 point

- ☐ `x = img.reshape((3,32*32))`
☐ `x = img.reshape((32*32,3))`
☒ `x = img.reshape((32*32*3,1))`
☐ `x = img.reshape((1,32*32,3))`

4. Consider the following random arrays `a` and `b`, and `c`: 1 point

```
a = np.random.randn(3,4) # a.shape = (3,4)
b = np.random.randn(1,4) # b.shape = (1,4)
c = a + b
```

What will be the shape of `c`?

- ☐ `c.shape = (3,1)`
☐ The computation cannot happen because it is not possible to broadcast more than one dimension.
☒ `c.shape = (3,4)`
☐ `c.shape = (1,4)`

5. Consider the two following random arrays `a` and `b`: 1 point

```
a = np.random.randn(1,3) # a.shape = (1,3)
b = np.random.randn(3,3) # b.shape = (3,3)
c = a * b
```

What will be the shape of `c`?

- ☐ The computation cannot happen because it is not possible to broadcast more than one dimension.
☐ `c.shape = (3,3)`
☐ The computation cannot happen because the sizes don't match.
☒ `c.shape = (1,3)`

6. Suppose you have n_x input features per example. Recall that $X = [x^{(1)} x^{(2)} \dots x^{(m)}]$. What is the dimension of X ? 1 point

- ☒ (n_x, m)
☐ $(m, 1)$
☐ $(1, m)$
☐ (m, n_x)

7. Consider the following array: 1 point

```
a = np.array([[2, 1], [1, 3]])
```

What is the result of $a * a$?

- ☐ $\begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}$
☐ $\begin{pmatrix} 4 & 2 \\ 2 & 6 \end{pmatrix}$
☐ The computation cannot happen because the sizes don't match. It's going to be an "Error!"
☒ $\begin{pmatrix} 5 & 5 \\ 5 & 10 \end{pmatrix}$

8. Consider the following code snippet: 1 point

```
a.shape = (4,3)
b.shape = (4,1)
```

```
for i in range(3):
```

```
    for j in range(4):
```

```
        c[i][j] = a[i][j] + b[j]
```

How do you vectorize this?

- ☐ `c = a.T + b`
☐ `c = a + b.T`
☐ `c = a + b`
☒ `c = a.T + b.T`

9. Consider the following arrays: 1 point

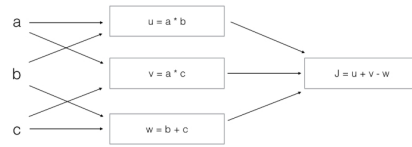
```
a = np.array([[1, 1], [1, -1]])
b = np.array([[2], [3]])
c = a + b
```

Which of the following arrays is stored in `c`?

- ☒ $\begin{pmatrix} 3 & 3 \\ 4 & 2 \end{pmatrix}$
☐ The computation cannot happen because the sizes don't match. It's going to be an "Error!"
☐ $\begin{pmatrix} 3 & 3 \\ 3 & 1 \\ 4 & 4 \\ 5 & 2 \end{pmatrix}$
☐ $\begin{pmatrix} 3 & 4 \\ 3 & 2 \end{pmatrix}$

10. Consider the following computation graph.

1 point



What is the output J ?

- ☒ $J = (a - 1) * (b + c)$
- ☐ $J = (b - 1) * (c + a)$
- ☐ $J = (c - 1) * (b + a)$
- ☐ $J = a * b + b * c + a * c$